

MADHAV BANJADE

Software Engineer

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PROFILE

Software engineer who builds features end-to-end — from UI to backend to data. Comfortable working directly in production codebases, debugging real issues, and integrating AI into live workflows.

EXPERIENCE

Full Stack Developer — HRN Management and Company

8 months

- Built and shipped production features end-to-end — including an automatic CV generation tool (Puppeteer/Handlebars) and internal dashboards — from requirement through deployment, under senior engineer guidance.
- Built and integrated REST APIs connecting frontend and backend services across employee, student, and attendance modules.
- Debugged a recurring PDF layout overflow in the CV generation pipeline by tracing it to CSS padding/min-height conflicts, then consolidated the stylesheet under shared custom properties — eliminating a dual-class-system conflict without touching the HTML structure.
- Worked directly with design and product teams to translate requirements into working features.

MERN Stack Intern — Nnine Solution

6 months, 2024

- Built reusable UI components inside a production MERN-stack codebase under senior developer mentorship.
- Integrated REST APIs connecting frontend interfaces with backend services.
- Completed a structured 6-month internship focused on engineering practices, code quality, and collaborative development workflows.

PROJECTS

HRN Nepal — System Dashboard — *Next.js · NestJS · TypeScript · MongoDB · Redis · AI*

- Contributed to a large-scale management system for an SSW visa consultancy, spanning employee records, student tracking, and attendance modules.
- Helped build automatic CV generation, document verification, and end-to-end visa processing workflows used across the consultancy's active caseload.

Fanout — Scaled Notification System — *NestJS · Nginx · Redis · BullMQ · Socket.IO · PostgreSQL · Docker* [Live demo](#)

- Designed a distributed notification system where stateless NestJS containers run behind an Nginx reverse proxy and load balancer, handling both HTTP and WebSocket traffic so throughput scales by adding containers rather than bigger servers.
- Used Socket.IO's Redis adapter to fan real-time events across all instances, so no client is pinned to a single server.
- Decoupled delivery from the request path with BullMQ/Redis queues and automatic retries, keeping the API fast under load; containerised with Docker.

Raag — DocuMind RAG — *NestJS · Qdrant · Groq · Jina · MongoDB · JWT · Render* [Live demo](#)

- Built a document Q&A app on an end-to-end retrieval-augmented generation pipeline: PDF, Word, and Excel files are parsed, split into overlapping chunks, embedded, and stored in a Qdrant vector database.
- Retrieved chunks are re-ranked with Jina and passed to an LLM on Groq that answers strictly from context, says so when nothing relevant is found, and returns the source chunks it used.
- Added JWT authentication and chat history with a lightweight vanilla JavaScript frontend; deployed the NestJS API on Render.

EDUCATION

Bachelor of Information Technology (Hons), Computing

Nov 2023 – Present

Phoenix College of Management & IT / Lincoln University College

SKILLS

Frontend: React, Next.js, TypeScript, TailwindCSS, Redux, Zustand, React Query, Framer Motion, Formik

Backend: NestJS, Node.js, Express, FastAPI, REST API, GraphQL, Socket.IO, BullMQ, JWT / OAuth

Databases: MongoDB, PostgreSQL, Redis, Prisma ORM, Qdrant (vector DB), Firebase

AI: RAG pipelines, embeddings, vector search, re-ranking, LLM integration (Groq, Jina)

Mobile: Flutter

Tools: Git, GitHub, Docker, Nginx, AWS, Render, Postman, Figma

Languages: English

ACHIEVEMENTS

- 3rd Place, Intra-College Hackathon — built a healthcare web app with an AI chatbot for remote-area access.